ARTIST FIELD GUIDE · NUKE 15.1

Put real air in your *shots*.

A friendly tour of two BlinkScript tools for physically based sky light and depth haze, with sliders that behave the way an artist expects.

EzAtmosScatter EzDepthHaze



The demo terrain with the Golden Hour look

EzAtmosScatter + EzDepthHaze

START HERE

1 The big picture

Two nodes, one job: putting the right amount of air between your camera and every pixel of terrain.

INPUTS
Your shot
A lat-long HDRI, the sun (in the HDRI or an Axis), the camera, and optionally the skydome.

→

THE WEATHER
EzAtmosScatter
What the air looks like in every direction: the sky colour, the sun glow, and the colour of deep haze.

→

THE DISTANCE
EzDepthHaze
Uses your depth to add haze, fog, shafts and shadows, the right amount at every pixel.

→

OUTPUT
Your comp
A hazed plate, with every pass as its own layer, ready to grade.

THE ONE IDEA TO REMEMBER

EzAtmosScatter is the weather: what colour the air is and where the sun sits. **EzDepthHaze is the distance**: how much of that air is in front of each pixel. Change the weather once, and every haze node fed by it follows.

INPUTS

hdri	A lat-long HDRI. It can stay empty with **self-lit** on.
sun	An Axis whose position points at the sun. Optional.
camera	The shot camera, for **analyze** and the skydome.
dome	The skydome geometry, if the HDRI sits on a dome in your comp.

plate	The render or plate, with its depth in a channel (`depth.Z`).
camera	The shot camera. Needed for perspective plates.
atmos	The EzAtmosScatter node. Without it, the haze uses a flat colour.

Every input can come through Dots and NoOps.

Left: EzAtmosScatter. Right: EzDepthHaze.

INSTALLING

No install. Import `groups/EzDepthHaze_group.nk` (or the EzAtmosScatter one) with **File › Import Script**, or paste its text into the node graph. The Group is self-contained, so it can live in a shot script or a toolset.

As a gizmo. Clone the repo into `~/.nuke/plugins/` and add `nuke.pluginAddPath('./plugins/EzDepthHaze')` to `~/.nuke/init.py` (and the same for EzAtmosScatter). Both nodes then appear in the Nodes menu.

INSIDE THIS GUIDE

2 Five-minute quick start
3 EzAtmosScatter, knob by knob
4 EzDepthHaze: the Easy tab
5 Looks
6 EzDepthHaze: the Advanced tab
7 Layers and the breakout
8 Recipes
9 Speed, farm and fixes

FIVE MINUTES

2 Quick start

From an empty script to a hazed plate. No physics degree required.

01 **Make the weather.** Create an EzAtmosScatter, plug the HDRI into `hdri` and the camera into `camera`. No HDRI? Tick **self-lit (no HDRI)**.

02 **Tell it the scale.** Set **units to metres** and **ground height** (in scene units), then press **analyze**. It sets the camera height, the haze distance and the sun source, and tells you what it set.

03 **Add the haze.** Create an EzDepthHaze under your plate. Connect `plate`, `camera` and the EzAtmosScatter into `atmos`.

04 **Check the depth.** On the Advanced tab, pick the **depth channel** and **depth type**. Then press **analyze** on the Easy tab: it copies the scale from the atmos node, spots 1/Z depth and finds a skydome behind the terrain.

05 **Pick a look.** Choose one from the **look** menu. It is fitted to your shot as soon as you pick it.

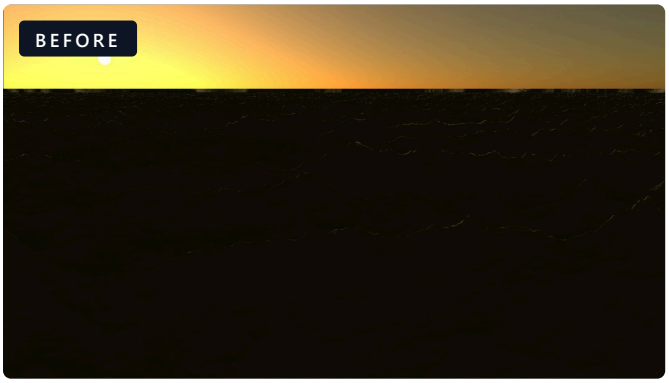
06 **Season to taste.** Push **haze**, **warmth**, **glow**, **shafts** and **fog**. The middle of every slider is the look as applied, so you can always find your way back.

STUDIO SETTINGS

Decimetre scenes: **units to metres** = **0.1**.
Camera Z depth: **depth type** = **Planar Z**.
Raw `1/Z` depth: **1/Z** (**analyze** spots it).
No position pass is needed. Depth plus the camera is enough, and the node rebuilds `P` and `N` itself.

WHERE DO I LOOK?

The main rgba shows whatever **view output** says. Every pass is also written to its own layer whatever you are viewing, so switching the view never changes what renders.



The raw terrain render.

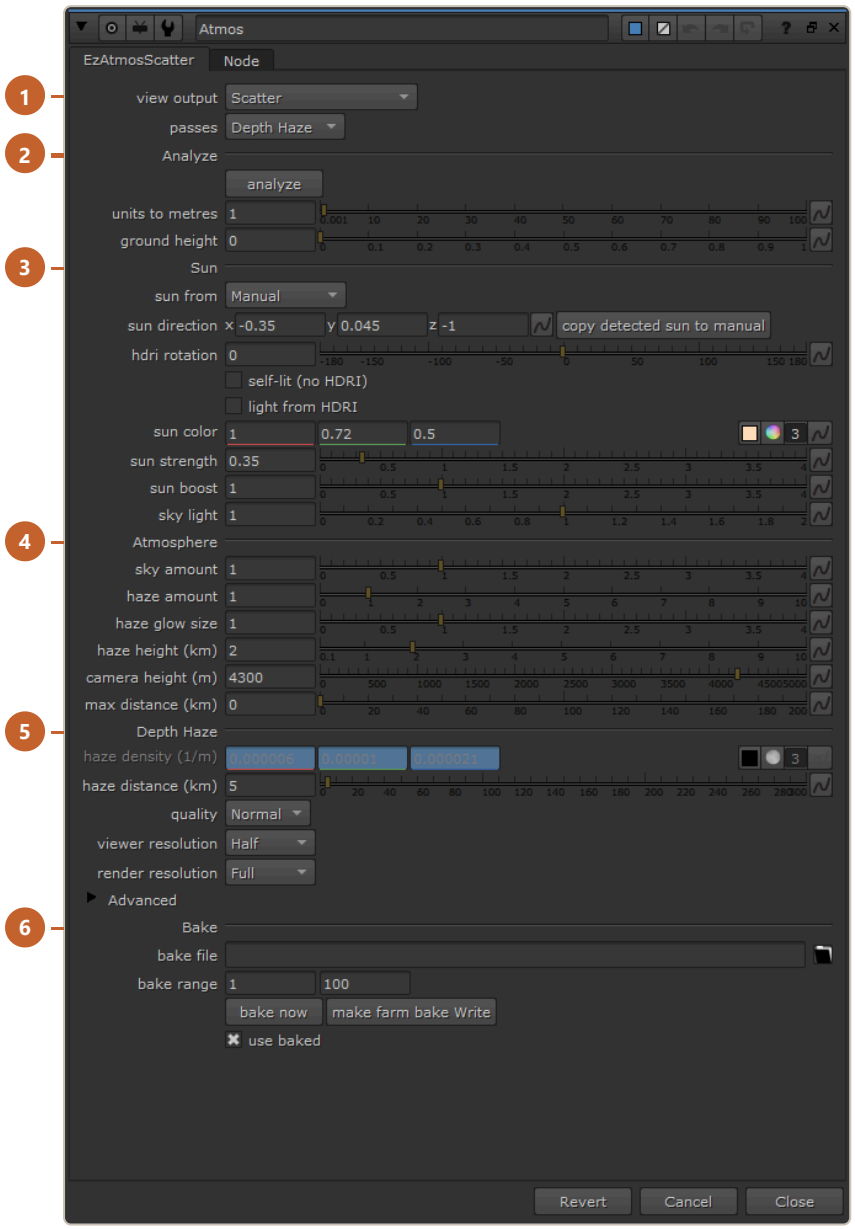


Golden Hour, picked from the look menu.

FOLLOW ALONG

The EzDepthHaze repo builds this demo for you. With both tools installed, run
`Nuke15.1.exe --nukex -i --gpu -t examples/build_demo.py <folder>`
It renders the mountains once and saves `ezdepthhaze_demo.nk` in that folder, ready for every step in this guide.

3 EzAtmosScatter



Six sections, top to bottom. Most shots only need 1 to 3.

1 Output
What you view, and which passes are built.

2 Analyze
The one-button setup from the scene's scale and camera.

3 Sun
Where the sun is, and how bright and what colour it is.

4 Atmosphere
How much air and haze there is, and how high it reaches.

5 Depth Haze
The numbers EzDepthHaze inherits, plus quality and speed.

6 Bake
Writes everything to EXRs for the farm.

1 OUTPUT

view output	What the main rgba shows: the scatter preview, sky, haze, haze colour and more.	
passes	**Depth Haze** gives haze around the camera out to *haze distance*, with no ground and no horizon line, which is what EzDepthHaze wants. **Atmosphere** gives a full sky down to the ground.	Depth Haze

2 ANALYZE

analyze	Sets *passes*, *camera height*, *haze distance* (out to the horizon) and the sun source. It warns if the sun is behind the camera or high in the sky.	
units to metres	Your scene scale. Use 0.1 for decimetres.	1
ground height	Ground level in scene units.	0

3 SUN

sun from	**Auto** finds the sun in the HDRI, **Axis** follows the `sun` input, and **Manual** uses *sun direction*.	Auto (HDRI)
sun direction	The manual sun, with +Y up. *copy detected sun to manual* fills it in from the HDRI.	
hdri rotation	Spins the lat-long.	0°
self-lit	No HDRI needed: the sky lights itself, and the sun comes from *sun color* and *sun strength*.	off
light from HDRI	When on, the sun's colour and brightness come from the image. When off, the two knobs below set them.	on
sun color · strength	Tint and multiplier on the sun.	
sun boost	Only for clipped suns (JPEGs, LDR captures).	1
sky light	How much of the HDRI sky lights the haze.	1

4 ATMOSPHERE

sky amount	Air density (Rayleigh). More gives a bluer, deeper sky.	1
haze amount	Haze density (Mie). More gives a milkier sky and a bigger sun glow.	1
haze glow size	Size of the glow around the sun.	1
haze height	How fast the haze thins with altitude. Keep it low for haze that hugs the ground.	1.2 km
camera height	Height above sea level. *analyze* sets it.	1 m
max distance	Stops the haze at a set distance instead of a full sky.	0 (off)

5 DEPTH HAZE, QUALITY AND SPEED

haze density	Read only: haze per metre at camera height. EzDepthHaze inherits it.	
haze distance	How far the Depth Haze passes reach. *analyze* sets it to the horizon.	5 km
quality	Samples per ray. Draft is about 4× faster than High.	Normal
viewer resolution	Size the haze is computed at in the GUI. The haze is smooth, so Half stays within about 2% of Full.	Half
render resolution	The same, for terminal and farm renders.	Full
Advanced	Physical constants and the sun detection settings. You rarely need these.	

6 BAKE

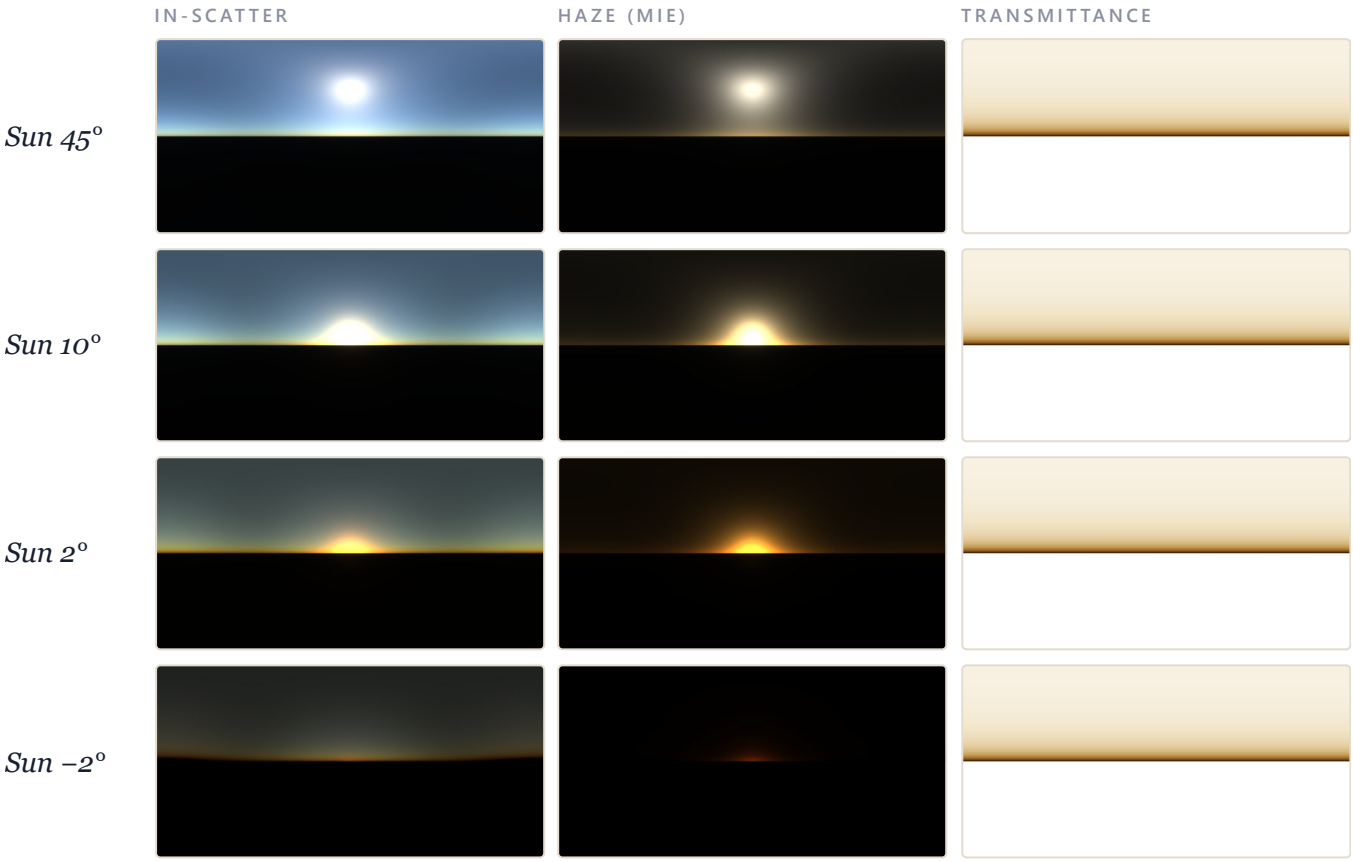
bake now	Renders every layer to EXRs over *bake range*. Set both frames the same to hold one frame for a sun that doesn't move.	
make farm bake Write	Adds a Write with the same path, so the bake runs as its own farm job.	
use baked	Reads the bake while it still matches the knobs. Otherwise the label says *bake out of date: live*.	on
bake file	Overrides where the bake goes. The default is `<script dir>/ezatmos_cache/…` or `$EZATMOS_CACHE_DIR`.	

SKYDOME IN THE COMP?

Plug the dome geometry into `dome` and the camera into `camera`. The node follows the dome's TransformGeos (animated ones too) and rebuilds the HDRI as the camera sees it, so the sun and haze line up with the dome. Leave *hdri rotation* at 0, and bake again after moving the dome.

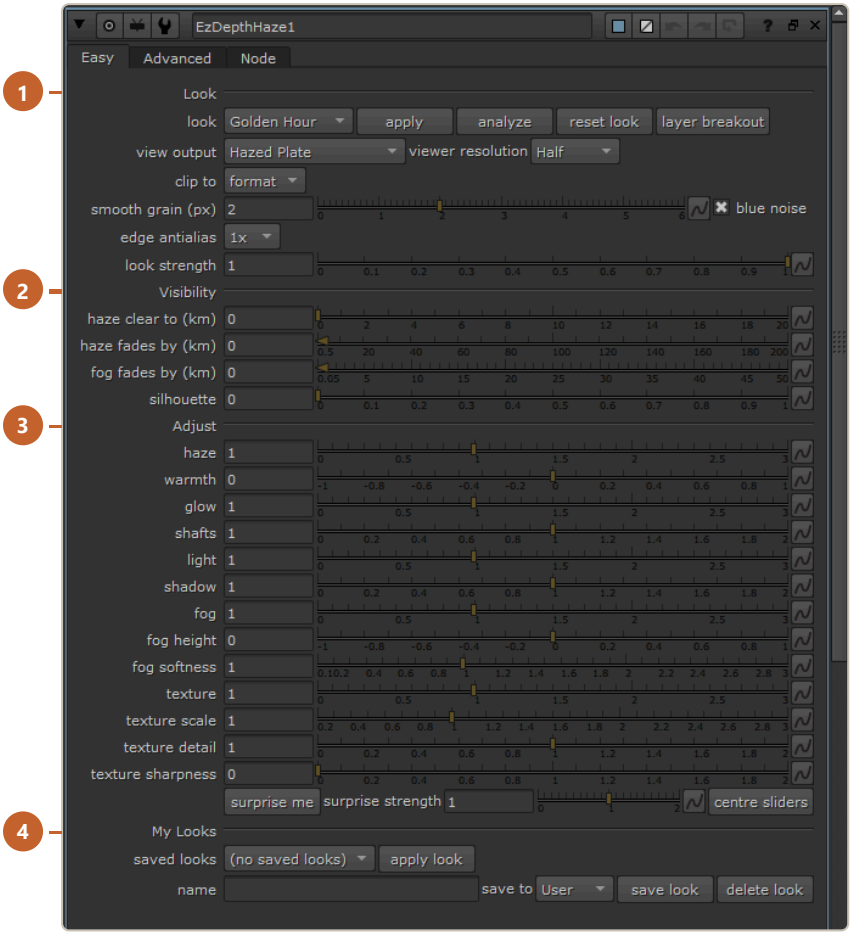
REFERENCE RENDERS

The same atmosphere as the sun sets. The glow around the sun tightens and warms, and the sky light fades.



Each tile is a full lat-long: the sky above the horizon, the ground below. Transmittance is white where nothing is lost and goes brown towards the horizon, where the air is thickest.

4 The Easy tab



The node opens here. Big friendly sliders, each moving several knobs.

1 Look
Pick a look, analyze the shot, set the output and quality.

2 Visibility
Where the air starts, where terrain fades, and the silhouette.

3 Adjust
Thirteen sliders, each relative to the look, plus surprise me.

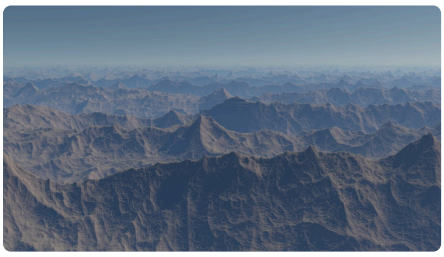
4 My Looks
Save, apply and delete your own looks.

1 LOOK

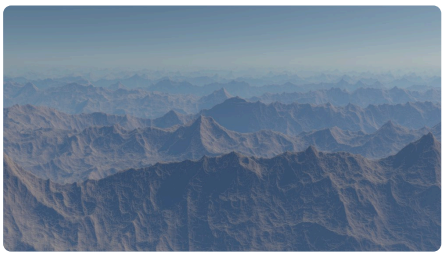
look	Picking one applies it straight away, fitted to your shot.	
apply	Applies it again, for example after connecting the camera or to undo your tweaks.	
analyze	Sets up the plate: units from atmos, 1/Z depth and the sky dome. It warns about a missing camera or an off-look sun.	
reset look	Puts every look knob back to the defaults. The plate settings stay.	
layer breakout	Rebuilds the result from its layers (see chapter 7).	
look strength	Goes from clear air (0) to the full look.	1
view output	Hazed Plate, Haze Transmittance or Haze In-scatter.	
viewer res.	Half or Quarter in the GUI, for about ¼ or 1⁄16 of the GPU memory. Renders always run at full size.	Half
clip to	Crops to the plate's format, its bounding box, or nothing.	format
smooth grain	Averages grain away, only between pixels at a similar depth.	2 px
blue noise	Spreads samples evenly, so any leftover grain is finer.	on
edge antialias	Smooths stair steps along ridge lines (Off to 3x).	1x

2 VISIBILITY

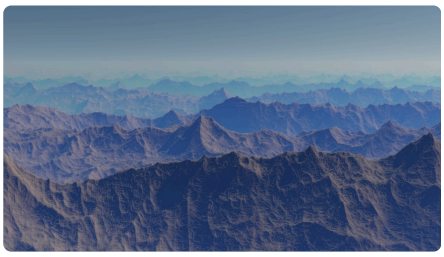
haze clear to (km)	Keeps the near air clear up to this distance.	
haze fades by (km)	The distance where terrain has faded into the haze. Typing a number sets the knobs behind it.	
fog fades by (km)	The same, through the fog.	
silhouette	Keeps the foreground's contrast while the far ranges close in faster, into flat layers the colour of the sky. Unlike more *haze*, it doesn't lift the whole frame.	0



Clear Day as applied.



Doubling *haze* lifts everything.



silhouette 0.7 layers the distance and keeps the foreground punchy.

3 ADJUST

Every slider is **relative to the look as applied**, and its middle is always the look. Push them to the edges to explore, then press **centre sliders** to come back.

SLIDER	WHAT YOU'LL SEE	KNOBS IT MOVES
haze	More or less air overall.	haze amount
warmth	Orange (+) or blue (−) haze.	haze tint, sky scatter
glow	A brighter halo around the sun.	sun scatter
shafts	Light rays. Past the middle you get full shadows, a sharper edge and brighter sunlit air.	shadow strength, penumbra, sun scatter
light	Sunlight on the terrain. Past the middle it switches surface light on.	surface light
shadow	Lighter (below the middle) or deeper (above) shadowed haze.	shadow fill, sky light
fog	A thicker, higher fog layer. It adds fog to looks that have none.	fog density, fog top
fog height	Moves the fog's top in steps of the terrain's height range.	fog top
fog softness	Low values give a crisp fog bank.	fog softness
texture	How much the air breaks into wisps.	noise amount, noise on haze
texture scale	The size of the wisps.	noise size
texture detail	Finer layers of wisps.	noise detail, roughness
texture sharpness	Soft blobs at the low end, crisp-edged patches at the high end.	noise sharpness

SURPRISE ME

Stuck? **surprise me** rolls random adjustments within tasteful limits. **surprise strength** goes from 0.5 (nudges) through 1 (usual) to 2 (wild).

EASY AND ADVANCED AGREE

The Advanced tab shows what Easy did. Edit a knob there and it becomes Easy's new starting point. **Loading or rendering a script never changes a knob.**

4 MY LOOKS

01 Type a **name**, pick **save to** **User** (just you) or **Studio** (everyone), and press **save look**.

02 Later, pick it in **saved looks** and press **apply look**. **delete look** removes it.

03 Saving over a look or deleting one keeps a dated copy in the folder's `.backup`, so nothing is lost. A pipeline can point the folders per show, and `$EZHAZE_USER_LOOKS` and `$EZHAZE_STUDIO_LOOKS` win when set.

STARTING POINTS

5 Looks

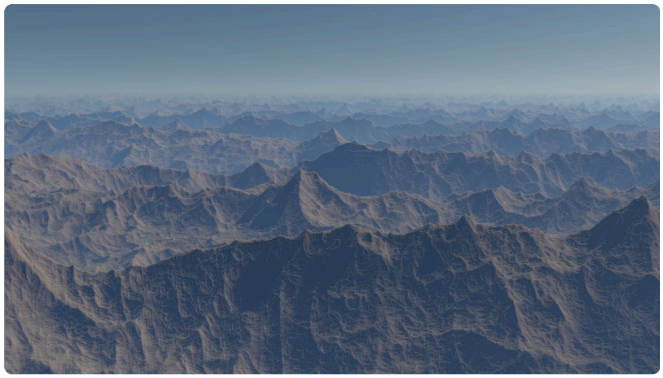
Five looks to start from, each fitted to your shot the moment you pick it.



Golden Hour

Warm haze, strong shafts and a light valley fog.

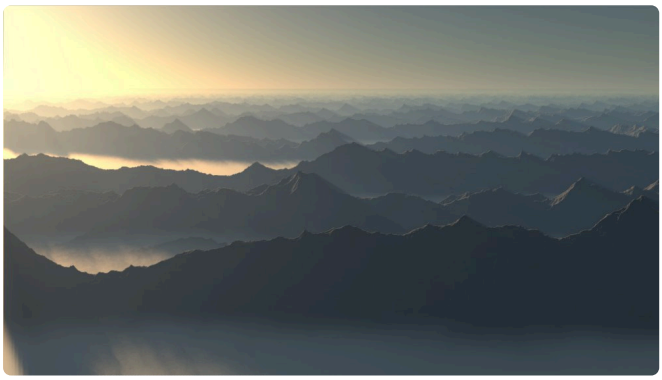
Sun: 4° up, 15° left, warm



Clear Day

Light blue aerial perspective, crisp shadows and no fog.

Sun: 35° up, 60° right, white



Valley Fog

A dense fog lake in the valleys, with clear air above.

Sun: 10° up, 30° left, slightly warm



Smoke Haze

Brown-orange haze with low contrast.

Sun: 8° up, 20° left, reddened



Overcast Mist

Grey-blue mist with drifting wisps and no sun shadows.

Sun: none

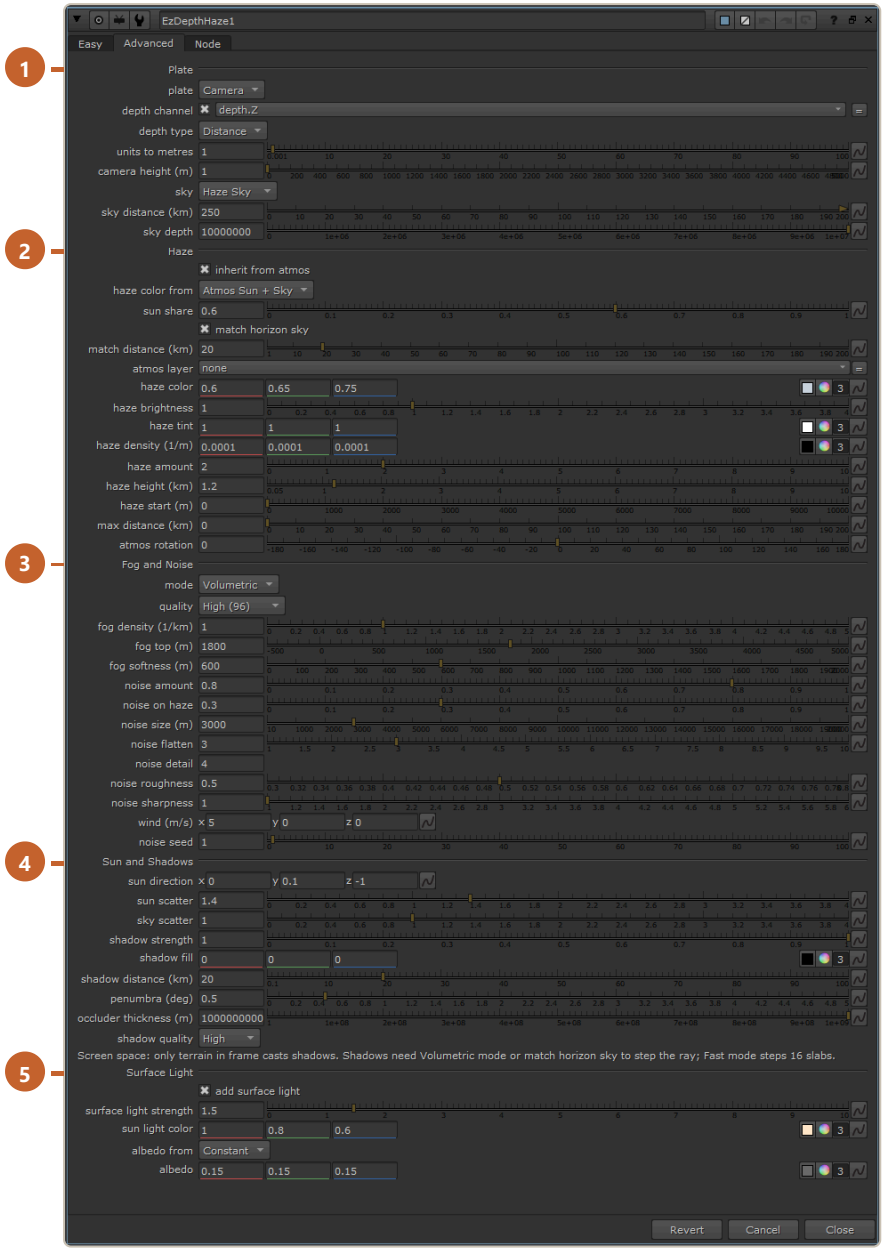
FITTED TO THE SHOT

Picking a look measures your depth and camera and rescales the haze, fog and noise to match. A 2 km shot gets the same *look* as the demo's 130 km, though not the same physics.

THE SUN POP-UP

If the sun is far from where the look wants it, you're asked whether to move it. Shafts and glow need the sun **low and in front of the camera**.

6 The Advanced tab



Every knob, set to whatever Easy chose. Come here to drive by hand.

1 Plate
Depth, units and sky handling.

2 Haze
Colour, density and inheriting from atmos.

3 Fog and Noise
The volumetric fog layer and its wisps.

4 Sun and Shadows
Shafts, shadow fill and the penumbra.

5 Surface Light
Sunlight falling on the terrain.

1 PLATE

plate	Camera (perspective) or Lat-long (360).
depth channel	Where the depth lives (`depth.Z`).
depth type	Distance, Planar Z or 1/Z.
units to metres	The scene scale (0.1 for decimetres).
camera height	For lat-long plates, in metres.
sky	Leave Sky or Haze Sky, for pixels with no depth.
sky distance	How far away hazed sky pixels are treated as being.
sky depth	Depth at or beyond which a pixel counts as sky.

2 HAZE

inherit from atmos	Takes the density, height and rotation from the atmos node.
haze color from	Atmos sun and sky, one atmos layer, or a flat colour.
sun share	How much of a single colour counts as sunlit.
match horizon sky	Far terrain fades into the sky behind it, over *match distance*.
color · brightness · tint	A flat colour, a multiplier, and a tint on all the haze light.
density · amount	Extinction per metre, and a multiplier on it.
haze height	How fast the haze thins with altitude.
haze start	The distance where the haze begins.
max distance	Caps the hazed distance (0 is off).
atmos rotation	Used when not inheriting.

3 FOG AND NOISE

mode	**Fast** is exact, in one step. **Volumetric** marches the ray, which fog and noise need.
quality	Samples per ray. More means less grain.
fog density	Thickness of the fog layer (1/km).
fog top	The altitude the fog fills up to. Set it a few hundred metres above the valley floors.
fog softness	How gradually the fog fades out at its top.
noise amount	How much the noise breaks up the fog.
noise on haze	How much of the noise also applies to the thin haze.
noise size	Size of the largest wisps and banks (m).
noise flatten	Squashes the noise into horizontal layers.
detail · roughness	Octaves of noise, and how strong the fine ones are.
noise sharpness	1 gives soft blobs; higher gives crisp-edged patches.
wind	Drifts the noise (m/s). It's fixed in world space, so it stays put when the camera moves.
noise seed	A different pattern.

4 SUN AND SHADOWS

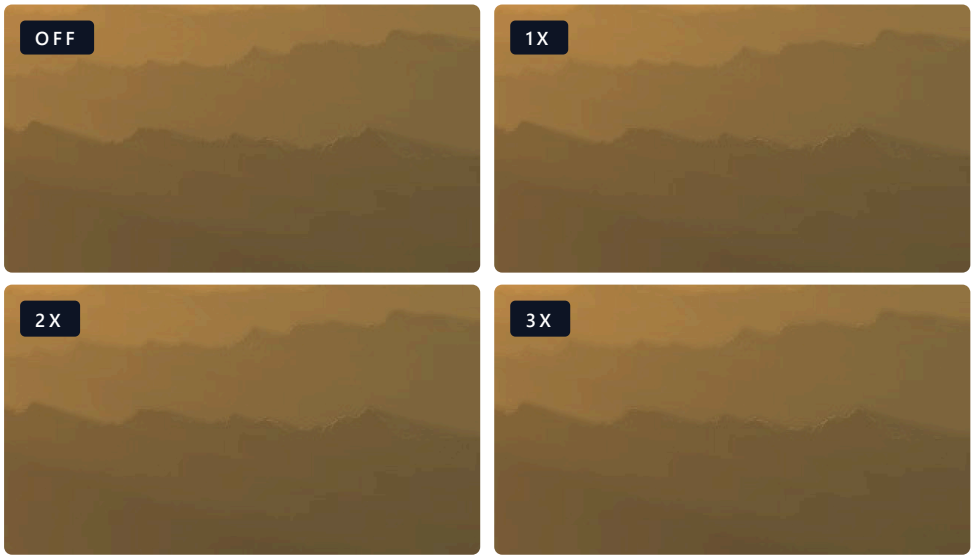
sun direction	Direction towards the sun, when not inheriting.	
sun scatter	Brightness of the sunlit haze (the shafts).	1
sky scatter	Brightness of the sky-lit haze (the haze in shadow).	1
shadow strength	Sunlight lost where terrain blocks the sun. 1 is physical and 0 is off.	1
shadow fill	Puts back a tinted share of the lost sunlight (artistic).	0
shadow distance	How far towards the sun each point looks for blockers.	20 km
penumbra	Shadow softness. The real sun is about 0.5° across.	0.5°
occluder thickness	How deep each surface is taken to be. Lower it for thin objects.	
shadow quality	Samples towards the sun. Raise it for finals.	Normal

GOOD TO KNOW

Shadows are found in screen space, so only terrain in frame casts them. They need Volumetric mode, or Fast mode with *match horizon sky* on.

5 SURFACE LIGHT

add surface light	Adds the sunlight falling on the terrain to the plate. Use it for photos and matte paintings, and leave it off for CG already lit by the same sun.	off
surface light strength	A multiplier.	1
sun light color	Colour and brightness of the sunlight on the terrain.	
albedo from	The plate's colour, or a constant *albedo*.	Plate



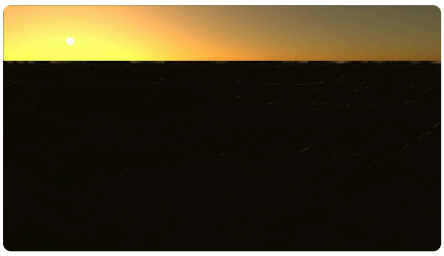
edge antialias up close on a ridge (Easy tab): Off shows stair steps, and 1x to 3x blend the haze across the edge over 1 to 3 pixels.

TAKE IT APART

7 Layers and the breakout

Everything the node does is written to its layers, so comp can take it apart and grade each piece.

```
result = (plate + surface_sunlight) × haze_transmittance + haze_sun + haze_sky
```



plate



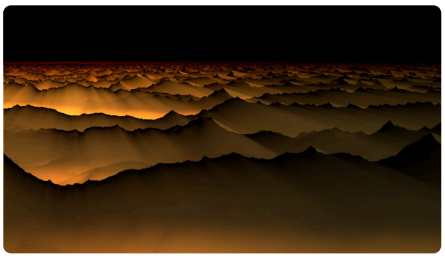
result



haze_sun



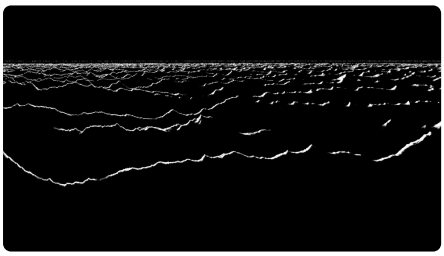
haze_sky



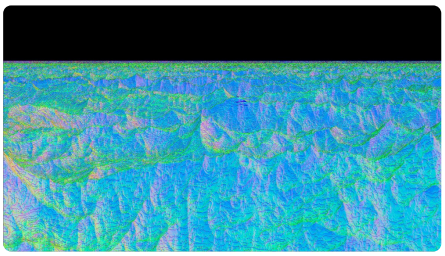
haze_shadow



haze_fog



surface_sun_visibility



N

haze_transmittance	How much of the plate survives the haze.
haze_inscatter	All the haze light (`haze_sun + haze_sky`).
haze_sun · sky	Sunlit haze (with shadows) and sky-lit haze.
haze_shadow	Sunlight the haze lost to shadows.

haze_fog · wisps	The fog's share of the light, and how much the noise changed it.
surface_sunlight	Sunlight on the terrain.
surface_sun_visibility	Its mask: 1 in sun, 0 in shadow.
P · N	World position (metres) and normal, rebuilt from depth.

THE LAYER BREAKOUT BUTTON

It builds a row of Dots off the node, with a Shuffle per layer merging down onto a trunk that starts from the plate. A difference check at the bottom should be black. That gives you a ready-made graph for grading each pass, and it's also where to add looks like distant desaturation or softening in comp.

COOKBOOK

8 Recipes

Common asks, and the sliders that answer them.

Low-lying valley fog

Fog filling the valleys, with the ridges standing clear.

1. Push **fog** up.
2. Pull **fog height** down until only the valley floors fill.
3. Lower **fog softness** for a crisp top, then add a touch of **texture**.

Golden hour shafts

Rays of light between the ridges.

1. Pick **Golden Hour** and say yes to the sun pop-up.
2. Push **shafts** past the middle, and add **glow** for the halo.
3. Grainy? See chapter 9.

Far ranges, flat and blue

Distant mountains stacking into soft layers.

1. Raise **silhouette** rather than **haze**.
2. Nudge **warmth** negative for a cooler distance.
3. Soften or desaturate the distance in comp, using `haze_transmittance` as the mask.

A clean foreground

Haze in the distance, crisp air up close.

1. Type a distance into **haze clear to**.
2. Air nearer than that stays clear.

Animated sunrise

A sun that moves and air that drifts.

1. Animate the sun on EzAtmosScatter (an Axis, or a keyed *sun direction* on Manual).
2. Set **wind** on EzDepthHaze so the wisps drift.

One look for the show

Every artist starting from the same place.

1. Dial it in, then **save look** to **Studio**.
2. Everyone sees it in **saved looks**, still fitted to each shot.

TROUBLESHOOTING

9 Speed, farm and fixes

GRAIN IN THE SUN HAZE

Keep **blue noise** on and **smooth grain** around 2. For finals, raise **quality** and **shadow quality**, and use **edge antialias** for crawling ridges.

GPU MEMORY TIGHT

Drop **viewer resolution** to Quarter. Renders still run at full size.

ON THE FARM

Terminal Nuke only uses the GPU with `--gpu` (`Nuke15.1.exe --gpu -x script.nk`). Without it, BlinkScript runs on the CPU, which still works but is slower. Each process compiles the kernels once (about 7 s on the GPU, 30 s on the CPU), so render several frames per task. Bake EzAtmosScatter to skip its kernels entirely.

QUICK FIXES

The haze is far too thick or thin	Check **units to metres** and **depth type**, then press **analyze** and **apply** the look again.
There's a hard line at the horizon	Turn on **match horizon sky**, and set **sky** to Haze Sky with a sensible **sky distance**.
The skydome is hazed like terrain	Press **analyze**, which sets **sky depth** to keep the dome out.
There are no shafts at all	The sun is behind the camera or high. Move it low and in front, which the sun pop-up does for you.
Everything looks washed out	Lower **haze**, or use **silhouette** for the distance instead.
The label says bake out of date	A knob changed since the bake. Press **bake now** again.
Lost after lots of tweaks	Press **centre sliders**, or **reset look** for a clean slate.

LAST TIP

The middle of every Easy slider is home. Push things to the extremes to find the edges, then back off until it feels right. Have fun with it.